ADITYA GHOSH

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY — **Atlanta, Georgia**
Master of Science in Computer Science — *Aug 2025 – May 2027*
Relevant Coursework: Machine Learning Systems, Deep Learning in Text Data, Deterministic Optimization — GPA: 3.87

VELLORE INSTITUTE OF TECHNOLOGY — **Vellore, India**
Bachelor of Technology in Computer Science and Engineering — *Sep 2021 – Jul 2025*
Specialization: Data Science — GPA: 3.96

EXPERIENCE

TRACTIAN — **Atlanta, Georgia**
Machine Learning Intern — *May 2026 – August 2026*

- Built a CatBoost churn prediction model using product usage, support interactions, and call-transcript embeddings, increasing identification of non-renewing customers three months before contract expiry from 34% to 68%.
- Engineered leakage-safe point-in-time features and walk-forward validation across 10+ classifiers, with an AUC-PR of 62% through SHAP analysis, permutation importance, and feature ablation.
- Built a Slack-based early-warning system that monitored customer usage signals and automatically notified Technical Account Managers (TAMs) when key metrics declined, enabling proactive intervention on at-risk accounts.
- Developed a RAG system over internal FAQ articles and customer-education video transcripts that retrieved relevant knowledge and surfaced specific video segments corresponding to customer questions.
- Developed an Iterative Corresponding Geometry (ICG) vision pipeline to estimate the pose and six joint angles of a 6-DOF robotic arm and integrated the predictions with its kinematic model to drive a real-time digital twin.

GEORGIA INSTITUTE OF TECHNOLOGY — **Atlanta, Georgia**
Graduate Assistant — *Dec 2025 – August 2026*

- Developed a cross-modal computer vision pipeline using optical microscopy images with SEM-derived labels to predict nanoscale additive-manufacturing quality, achieving 97.8% grouped accuracy for structure-level classification.
- Built area-level classification models to identify localized under- and over-curing across print regions, achieving 86.2% average accuracy and up to 93.1% accuracy on individual region classes.
- Demonstrated transfer learning across different printed geometries, showing that features learned from cylindrical structures generalized effectively to previously unseen woodpile structures. [Paper submitted to Journal Additive Manufacturing]

FEOLA LAB, EMORY UNIVERSITY — **Atlanta, Georgia**
Student Researcher — *Jan 2026 – May 2026*

- Engineered a multi-modal 9-class ophthalmic image segmentation model using a ResNet50 backbone across thousands of fundus and OCT images spanning vascular, glaucoma, diabetic retinopathy (DR), retinal structure, AMD, DME, and IRF fluid datasets; implemented weighted cross-entropy and soft Dice loss, achieving 0.81–0.99 Dice scores.
- Applied warm-start transfer learning for cross-dataset fine-tuning and managed experiment tracking, hyperparameter tuning, and performance logging using Weights & Biases (W&B).

INDIAN STATISTICAL INSTITUTE — **Kolkata, India**
Research Intern — *Jun 2024 – Jul 2024*

- Engineered a low-resource training pipeline by fine-tuning a 4-bit quantized large language model (LLM) Mistral-7B model with LoRA adapters, reducing training time by 50% while preserving model quality and reducing memory usage.
- Expanded query understanding by converting short queries into detailed questions and evaluated semantic similarity of model outputs using SBERT embeddings, achieving an average cosine similarity of 0.7218 on a held-out 10% test set.

PROJECTS

Natural Language Inference–Based Hallucination Detection (Data4Good 2025)

- Developed an ensemble solution consisting of a finetuned DeBERTa MNLI model and Azure OpenAI-based gpt-5-mini LLM-as-a-judge approach to detect factual consistency, contradiction, and irrelevance in LLM-generated responses.
- Regional champions (1st place East region out of 1015 teams) of the national Data4Good Case Challenge 2025 with a final 96.5% accuracy score.

SKILLS

Languages: Python, R, SQL, C++
Tools: TensorFlow, PyTorch, Keras, Scikit-learn, Tableau, Spark, Transformer-based Models, LangChain, LangGraph
Deployment: Github, Git, AWS, Docker, GCP, REST API, Databricks, Linux, Version Control, Kubernetes, Streamlit